Assignment - 1

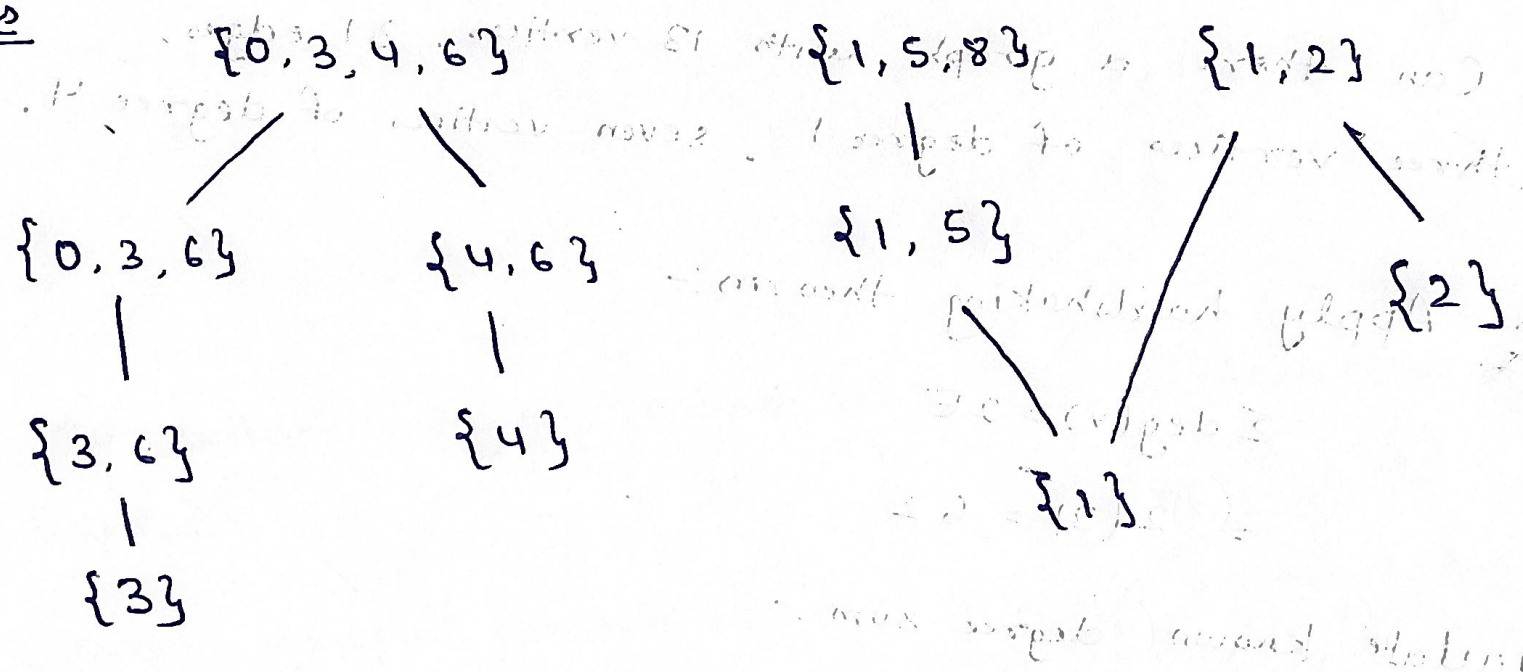
Discrete Mathematics

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Q1 Given S = {{1}, {2}, {3}, {4}, {1,2}, {1,5}, {3,6}, {4,6} {0,3,6}, {1,5,8}, {0,3,4,6}} and the relation R, being a subset of relation. Draw Hasse diagram.

Ans



Q2 Total vertices = 24, total edges = 30, five vertices have degree 4, seven pendant vertices have degree 1. Seven vertices have degree 2, remaining vertices have degree 3 or 4. find no. of vertices of degree 4.

Ans Handshaking theorm :-

$$\Sigma \deg(V) = 2E$$

$$= 2(30) = 60$$

remaining vertices

$$24 - (5+7+7) = 5$$

let x = no. of remaing vertices of degre 4.

then

$5-x$ have degree 3.

Equation :-

$$5(4) + 7(1) + 7(2) + 4x + 3(5-x) = 60$$

$$50 + x = 60$$

$$x = 4$$

therefore,

total vertices of degree 4 : $5+4 = 9$ Vertices

Q.3 Can there a graph with 13 vertices, 31 edges, three vertices of degree 1, seven vertices of degree 4.

Ans Apply handshaking theorm :-

$$\Sigma \deg(v) = 2E$$

$$= 2(31) = 62$$

Calculate known degree sum :-

$$3(1) + 7(4) = 31$$

Remaining = 31

Max. possible degree = $13-1 = 12$

Only possible combination :-

(7, 12, 12), (8, 11, 12), (9, 10, 12), (9, 11, 11), (10, 10, 11)

Every single from these requires at least 4 edges to connect to the degre-1 vertices, which is impossible.

Therefore, no such graph exists

Q4 no. of edges = 35, every vertex has degree atleast 3, find max. no. of vertices.

Ans $\Sigma deg(v) = 2E$ [handshaking theorm]

$2(35) = 70$

max. degree condition

$3V \leq 70$

$V \leq 23.33$

$\boxed{V_{max} = 23}$

This maximum is achievable

4, 3, 3, . . . 3

Ans → 23 vertices.

Q-5 for what integer m and n is $K_{m,n}$ Hamiltonian? Explain.

Ans Biparite graph consists of two disjoint vertex sets :-

- Set A countains m vertices.
- Set B contains n vertices

~~Every~~ Condition!

Both sets must have equal vertices

$m = n$

Condition 2

Each set countain at least two elements

$m = n \geq 2$

Therefore,

$K_{m,n}$ is hamiltonian if $m = n \geq 2$.

Q 6 Given $E = 3V - 6$, prove that every region of the connected planar graph G is a triangle.

Ans Euler's formula:-

for connected planar graph

$$V - E + F = 2$$

Substitute $E = 3V - 6$

$$F = 2V - 4$$

Apply face- degre formula

$$\Sigma \deg(R) = 2E$$

$$= 2(3V - 6)$$

$$= 6V - 12$$

So,

$$3F = 6V - 12$$

Hence

$$\Sigma \deg(R) = 3F$$

Each region has degree 3

hence,

every region is triangular.

Q-7 Graph Coloring problem

a) A zoo plans to remodel by removing cages and place in large enclosed area. Any animal that can't live together must be in diff. enclosure. The zoo will determine min. no. of enclosure needed.

Ans

Vertices (Nodes): Each individual animal (or species)

Edges (Connections): Edge is drawn between two vertices if corresponding animals can't live together.

Colours: The distinct enclosure where animals will be placed.

The minimum number of colors required to color the graph, corresponds to min. number of peaceful enclosure.

(b) English dept. is scheduling. Each student has made a list of course. Dept. would like to make schedule so that every student enroll in their desired course

Ans Vertices (Node): Each course offered by english department.

Edges (Connections): Edge is drawn between two vertices if atleast one student has enrolled in both courses.

Colours: The distinct time slots in schedule.

min no. of color required is min no. of time slots.